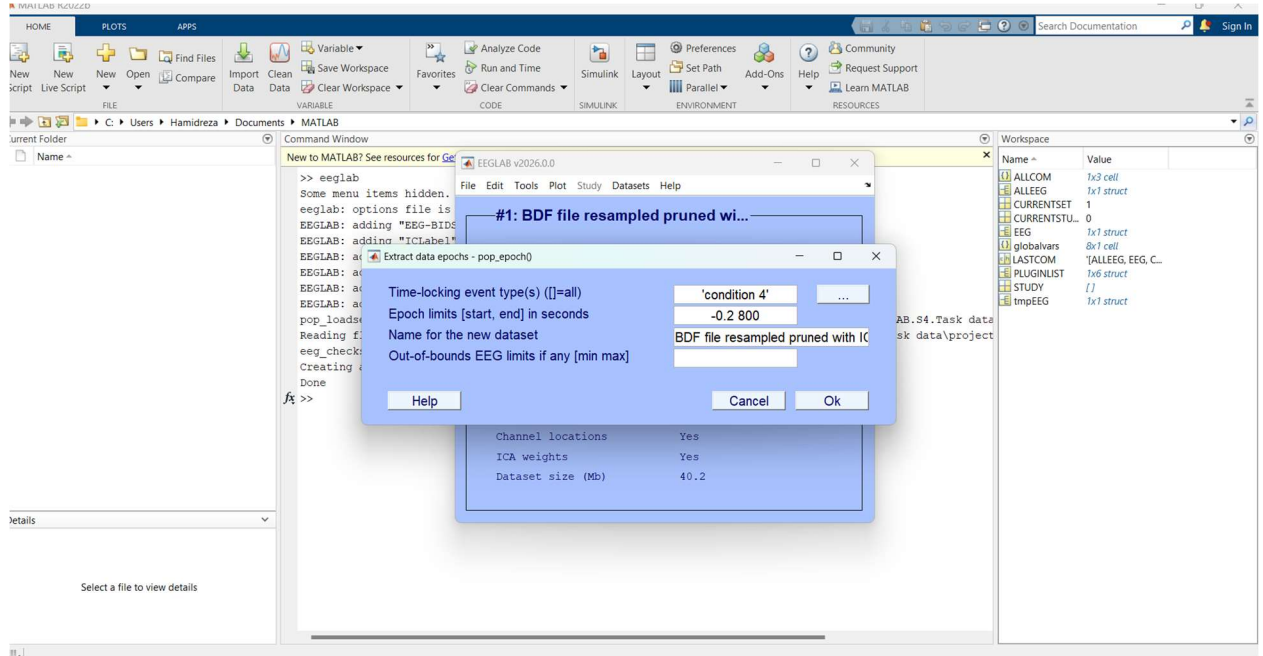


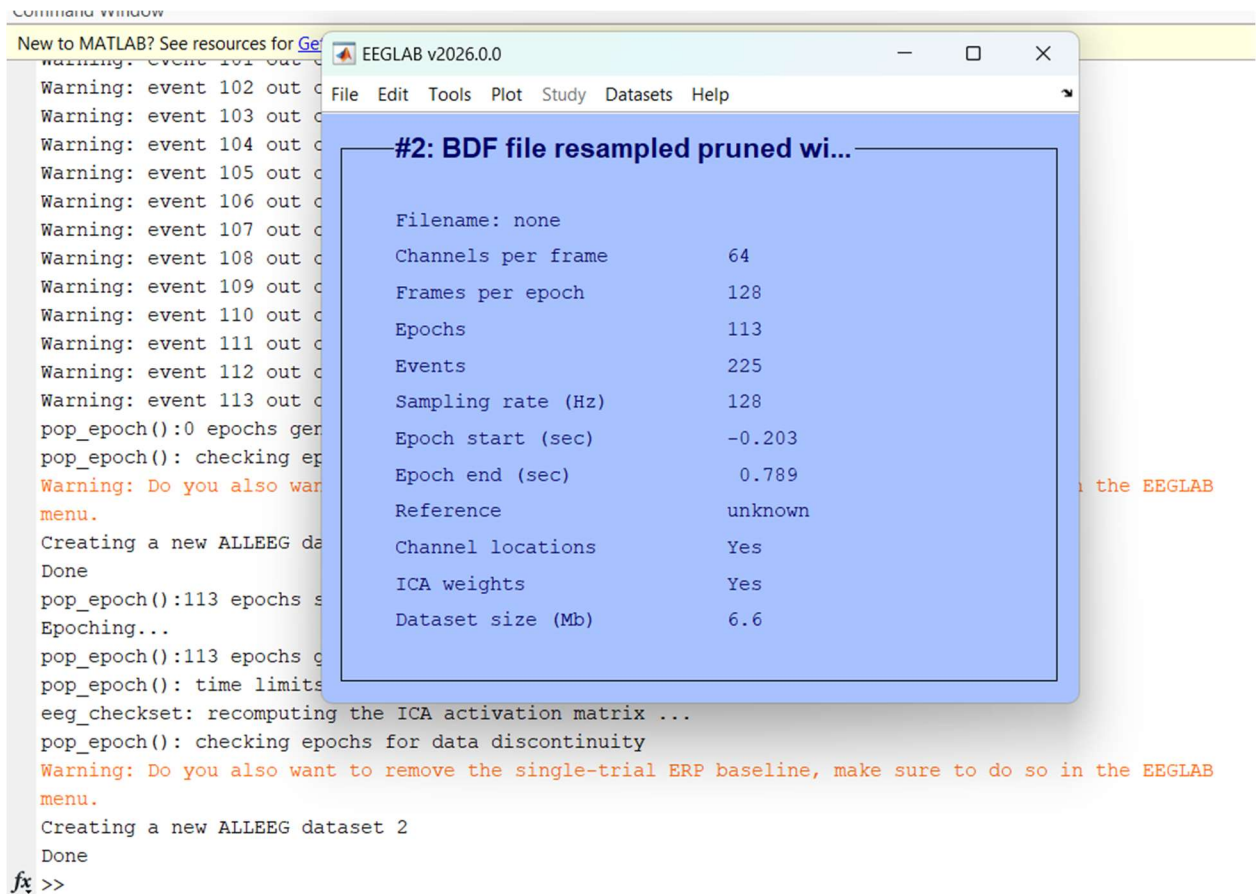
Assignment4

Name: Hamid Reza Ghahramani

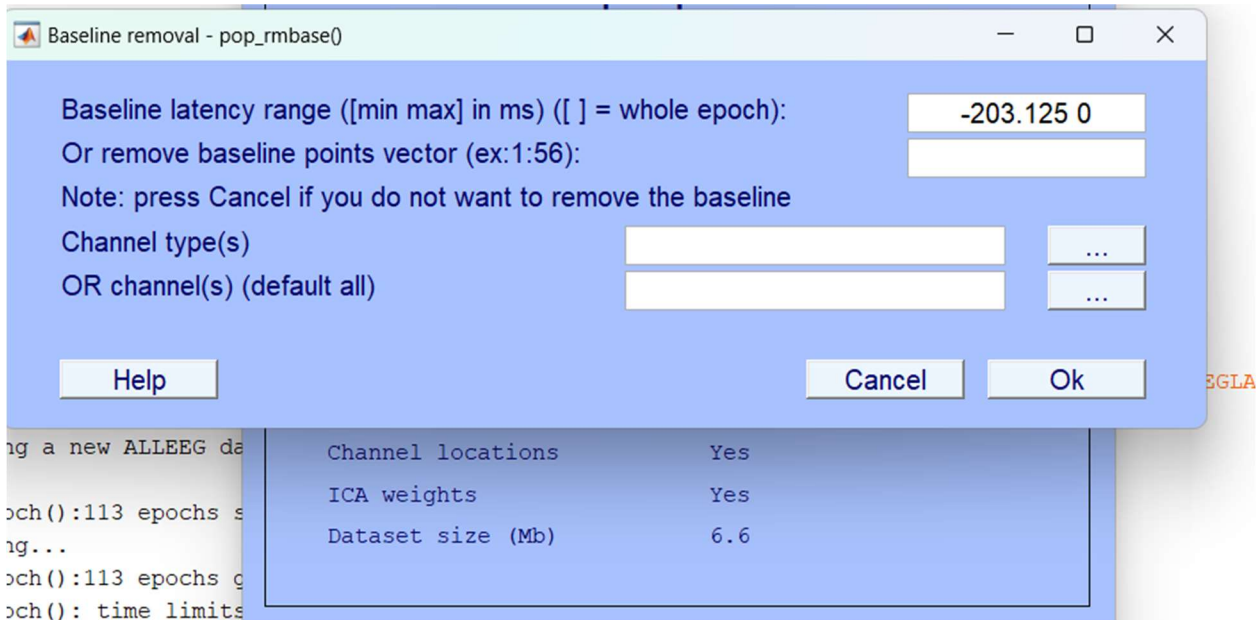
Using project04_preprocessed.set data set:

1. Extract epochs within the [-0.2 800] time range for event code = condition 4 and take screenshots of the steps.

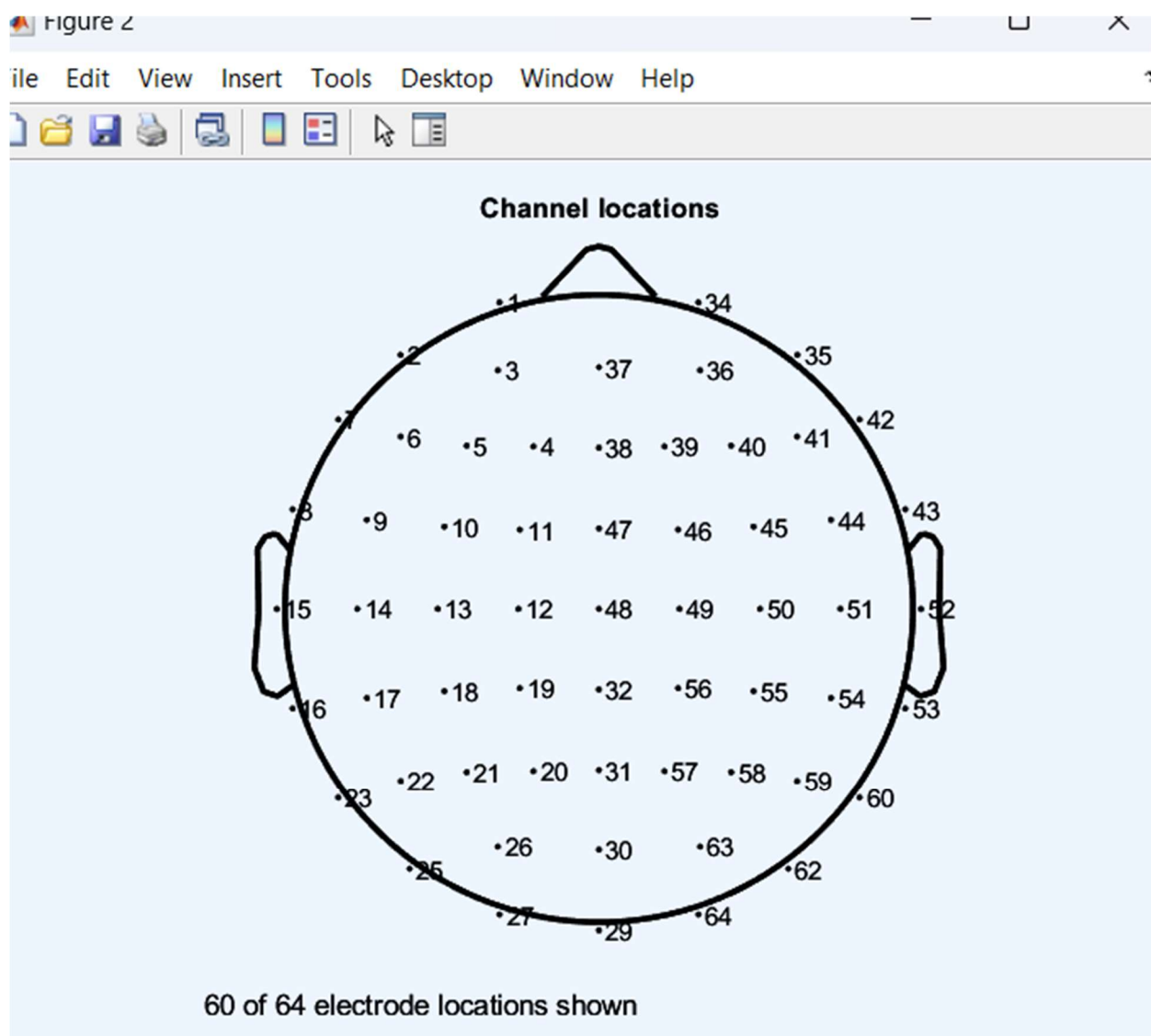


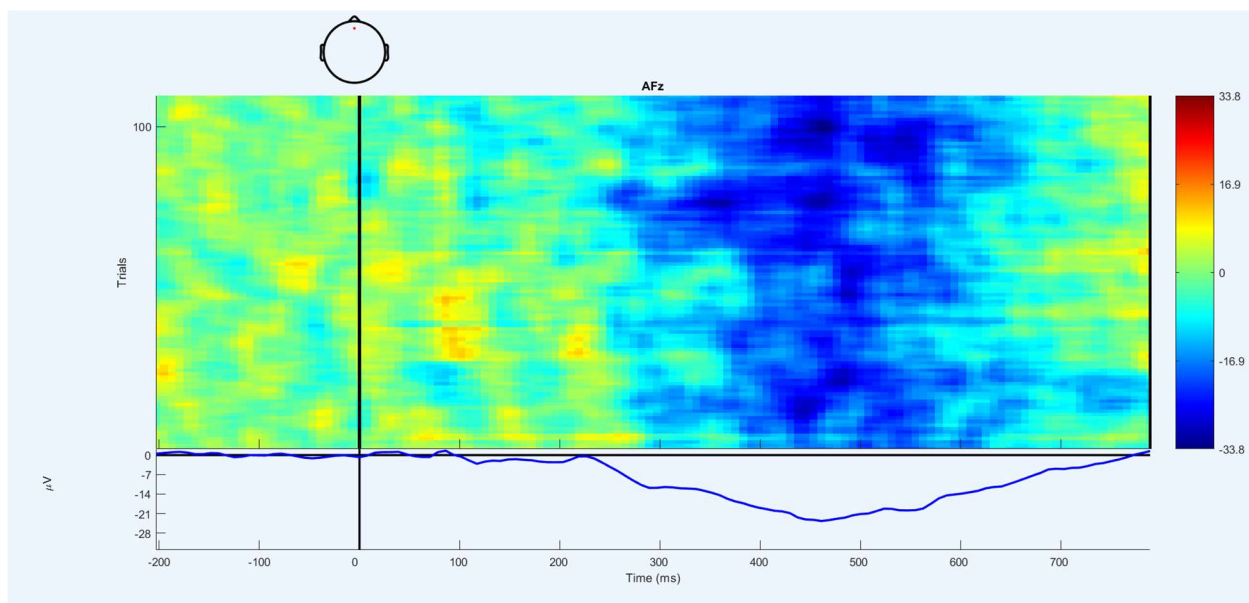


2. Execute baseline correction and take screenshots of the steps.

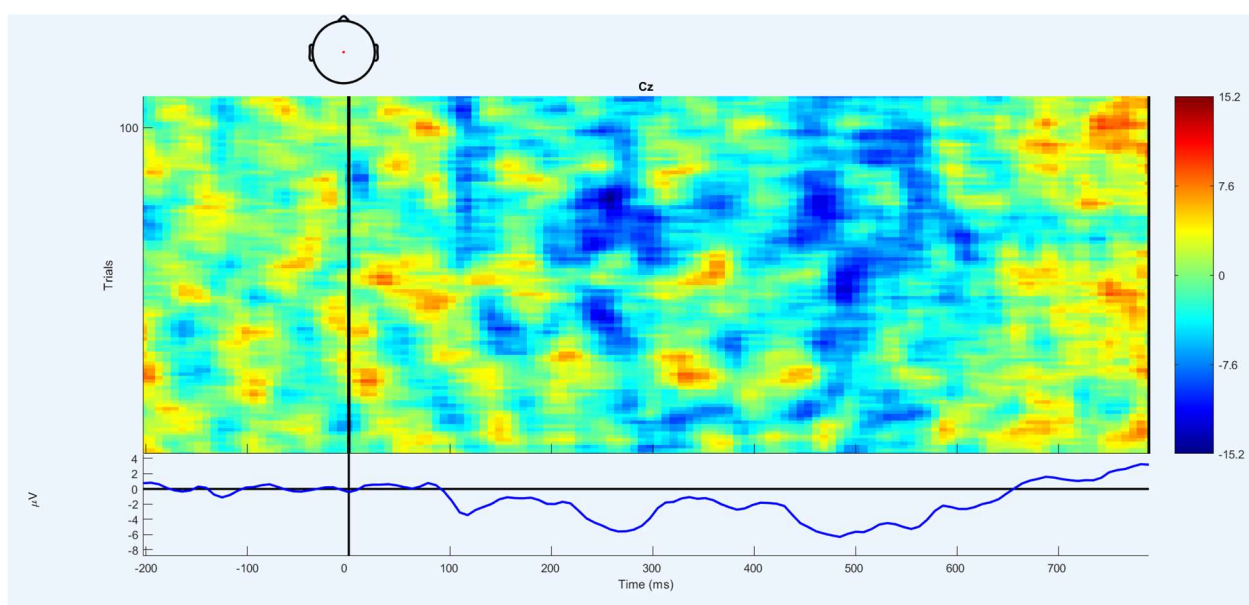


3. Plot ERP images of one frontal, one central, and one parietal channel and take screenshots of the steps and explain the components you see in each one.

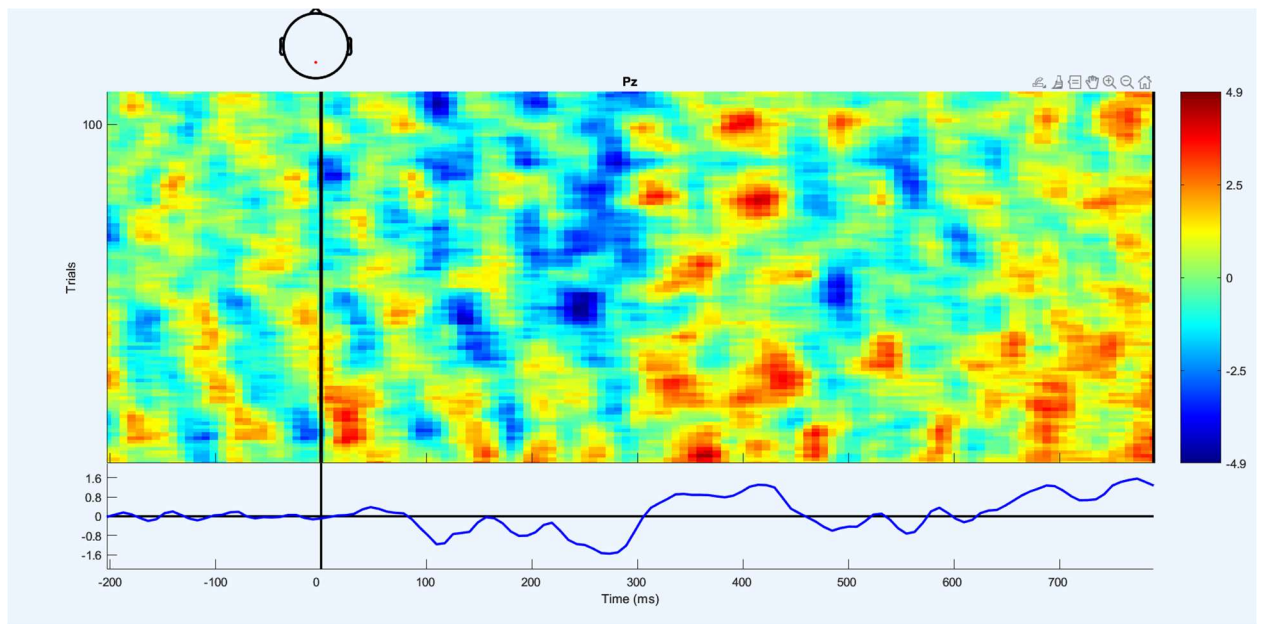




Frontal



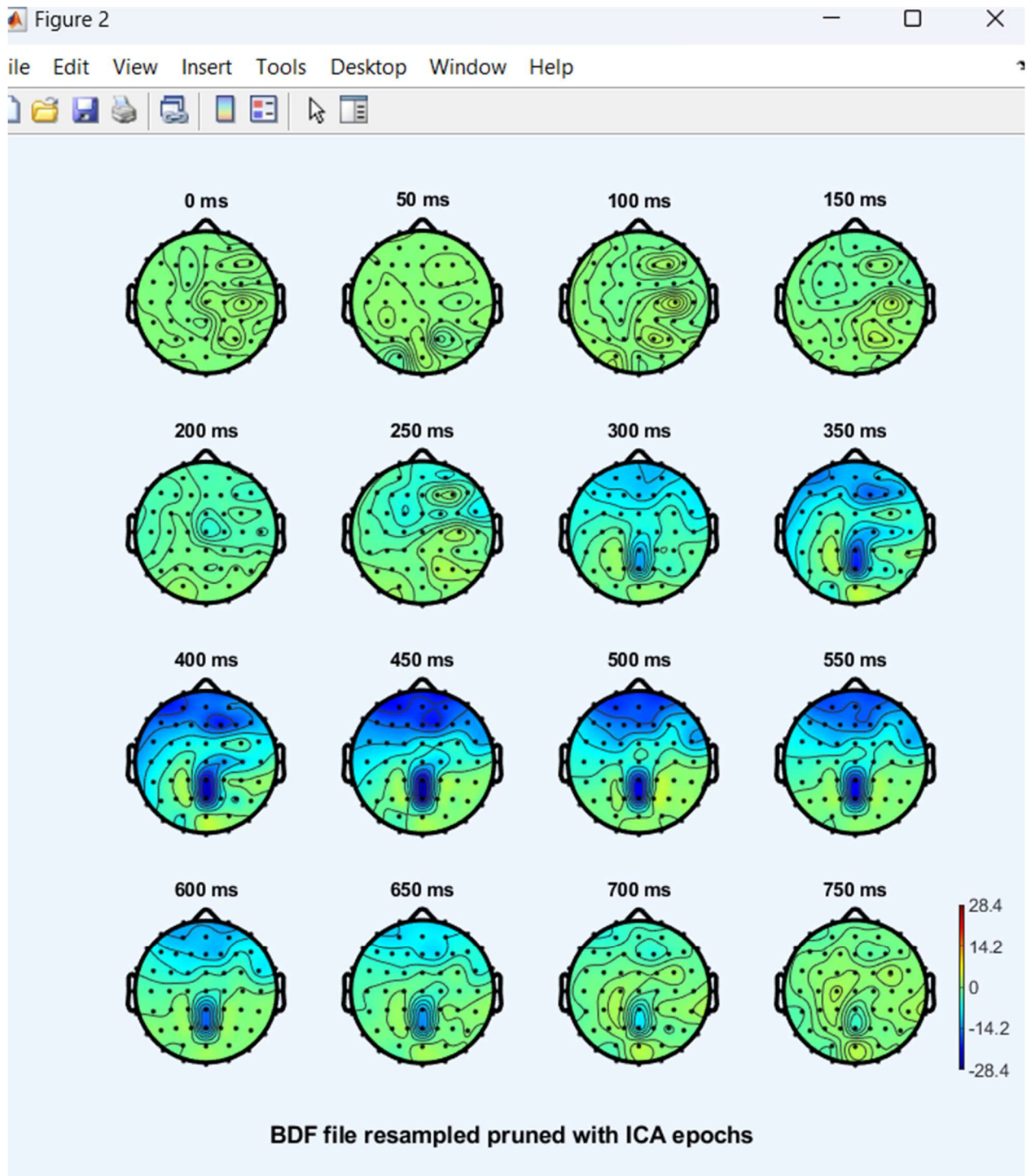
Central



Parietal

The ERP image shows greater activity over the parietal regions, where cognitive-related components are more clearly observed. In contrast, these effects appear weaker and less pronounced over the frontal regions.

4. Plot ERP map series in 2D in [0:50:750] time range and take screenshots of the steps.



- Repeat all the steps for event code = condition 2. What do you notice? (Hint: You need to compare the conditions)

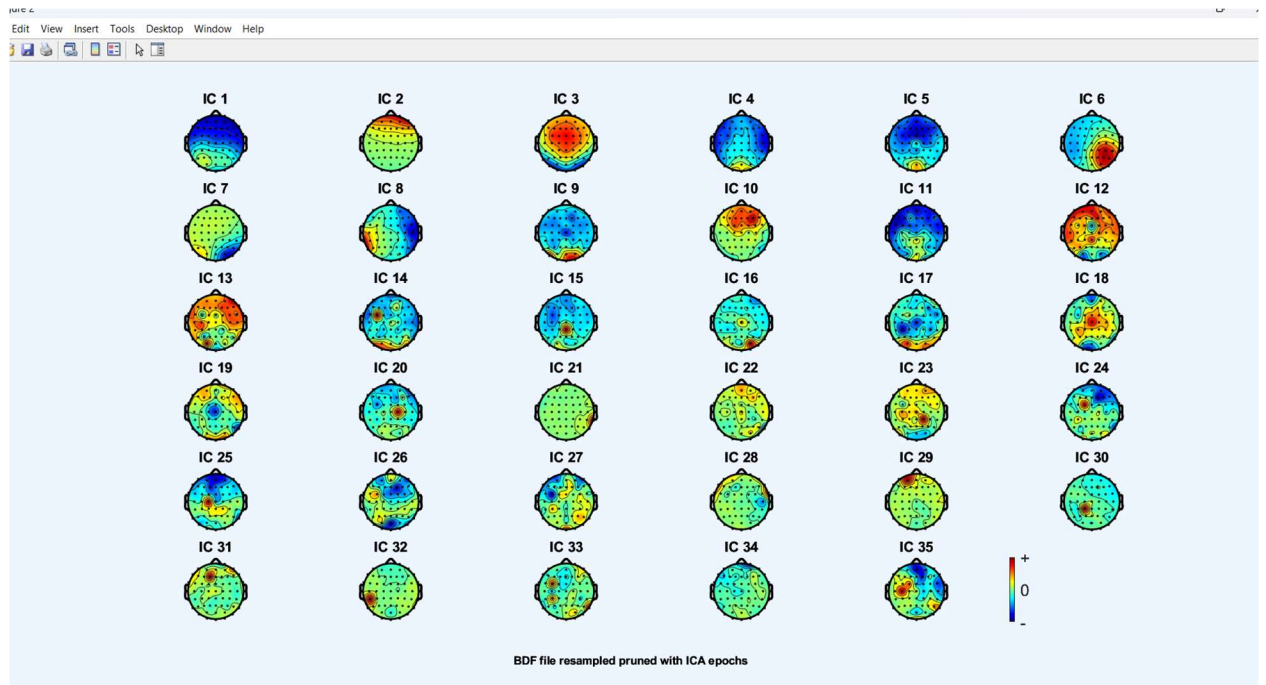
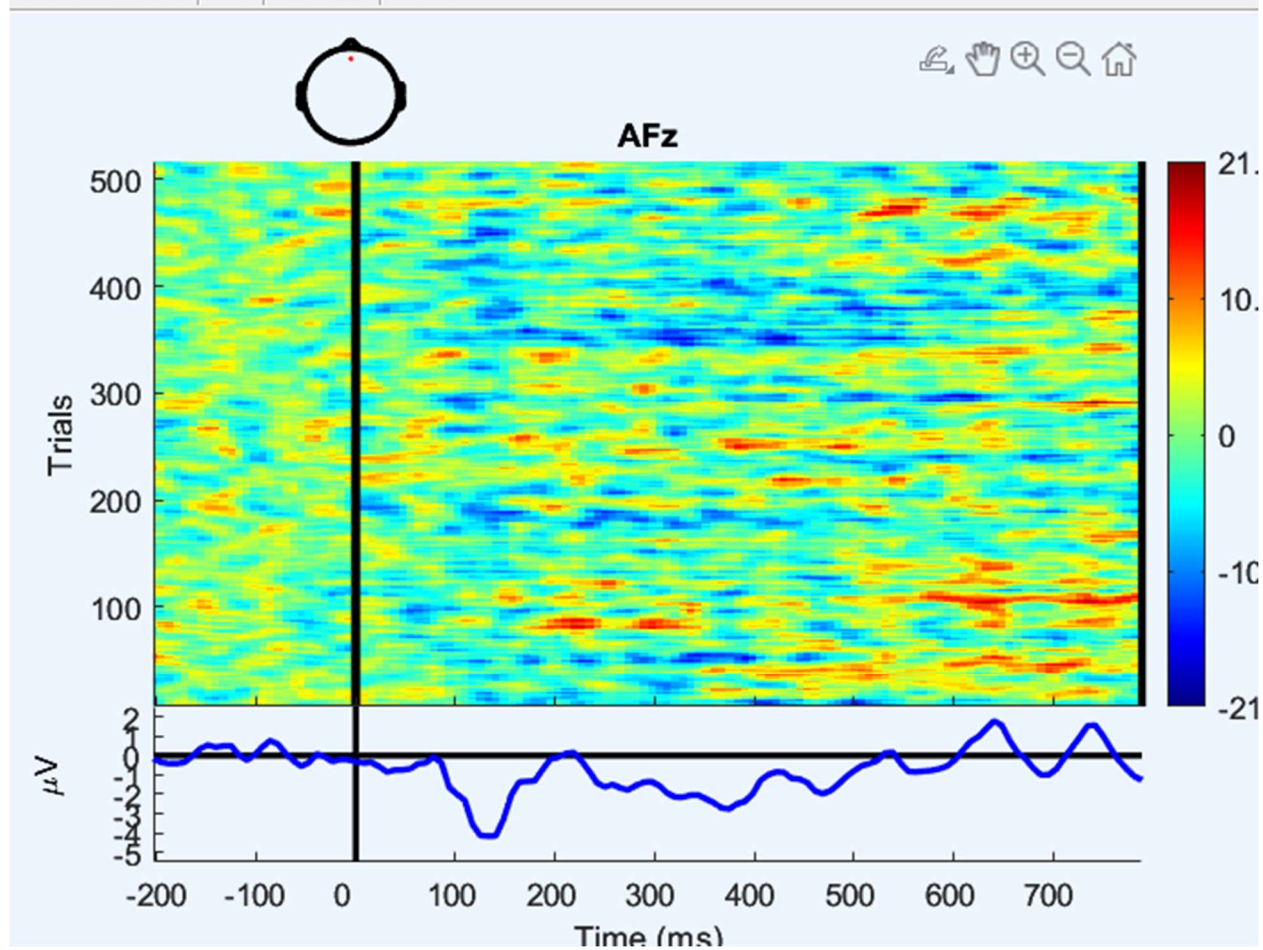
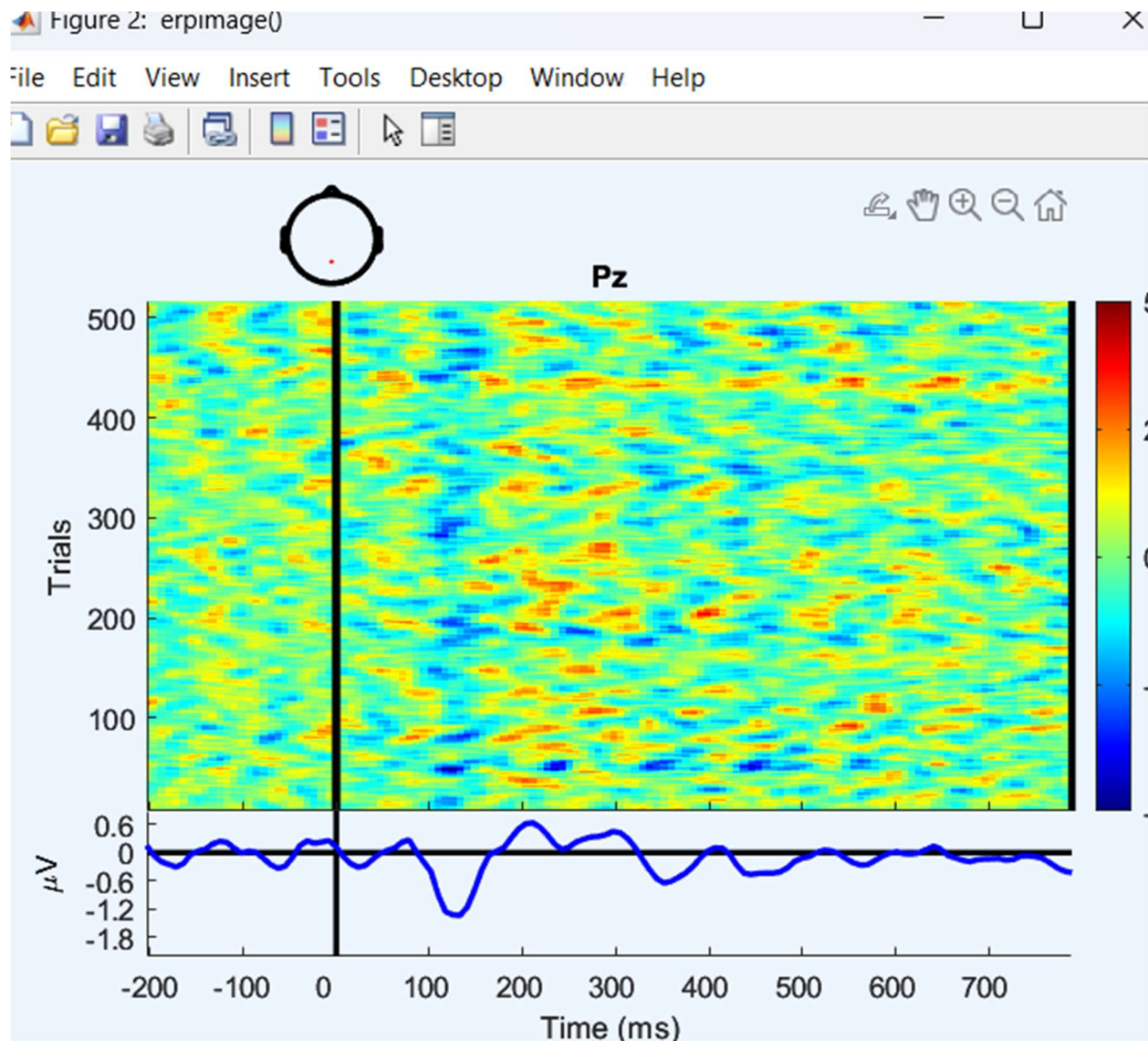


Figure 2: erpimage()

File Edit View Insert Tools Desktop Window Help





Compared with Condition 4, Condition 2 shows greater cognitive-related activity over the frontal region. This effect is more apparent when comparing the AFz and Pz channel locations, suggesting a relatively stronger expression of the ERP components in the frontal region in Condition 2 than in Condition 4.